

Twiceheard

A voice agent answers the clinic's phone, takes the patient's details, and books the appointment. Every critical value is heard twice, and the chart the clinic receives says which values were verified and which were not.

The name is the claim. One hearing is the live conversation. The second is the recording, transcribed again on its own with a confidence score on every word. A line is green only when both agree.

The problem

A clinic can let a voice agent answer the phone. What it cannot do is act on a name, a date of birth or a phone number that only a model says it heard.

Intake is where a visit starts and where it goes wrong. A transposed digit in a phone number means the reminder never arrives. A missed allergy is a clinical risk. A drug name heard as a similar one is worse than no drug name at all.

Voice agents today hand over a transcript and a summary. Both are the model's account of the call. If the model was wrong, nothing in the artefact says so, and the front desk has no way to tell a confident mistake from a correct answer.

The failure is silent, which is what makes it expensive.

What Twiceheard does

On the call. The agent collects the reason for the visit, name, date of birth, phone number, medications and allergies, then offers real open times from the clinic's diary and books one.

Every value is read back. The sentence is built by the server, not by the model, so digits are spoken in the groups people use and a date never comes out in the wrong order.

After the call. The chart is rebuilt from the session record rather than from the model's report, and the recording is transcribed a second time with per-word confidence.

The clinic receives a graded chart. Each line is **verified** or **to check**, and every line short of verified carries the reason in plain words.

How the verification works

Read back	The server composes the sentence and the agent is told to say it exactly.
A yes binds	Only an answer to that readback confirms that value. One yes cannot confirm two values.
Rebuilt, not reported	After the call the chart is replayed from the session record. A confirmation the record does not support is downgraded, whatever the model said at the time.
Heard again	The stored recording is transcribed a second time, in batch, with the speakers on separate channels and a confidence score on every word.
Green means both	A field is verified only when the conversation and the recording agree. Anything else is handed to the desk with the reason.

The rules are enforced in code and each one is pinned by a test that fails without it.

Built on

AssemblyAI Voice Agent API. The live call: streaming speech, turn detection, LLM routing, voice output and JSON schema tool calling. A stored agent answers the phone line; a browser call runs the same prompt and the same tools inline.

AssemblyAI Universal-3.5 Pro. The second hearing. Pre-recorded, multichannel, per-word confidence, transcript deleted as soon as it has been scored.

The rest. Next.js on Vercel, pinned next to the voice platform. Upstash Redis for intakes, slot claims and call records. RxNorm for medication names. Twilio SIP for the phone line.

Nothing clinical is left to the model. Every tool argument is validated on the server, a booking needs three confirmed fields, and an emergency is handed to a person.

What was measured

Test	Result
Unit tests, with 100% statements, branches, functions and lines enforced	483 pass
Scripted evaluation cases across grading, the recording, confidence and readback	26 pass
Browser tests, including a whole call with a synthetic microphone	27 pass
Median gap between the caller finishing and the agent speaking, on a live call	3.4 s
This application's own tool handlers, median, in that gap	20 ms
Two callers racing for the same appointment slot, in production	one booking stood

The latency is the platform's turn, not this application's work, and it is slower than the paid products in this market. It is listed here because it is true.

Market scope

The addressable unit is one outpatient practice phone line.

Counted	Figure	Source and date
Private hospitals empanelled under Ayushman Bharat, all running outpatient departments	16,746	IBEF, 28 February 2026
Public hospitals empanelled under the same scheme	19,483	IBEF, 28 February 2026
Registered allopathic doctors in India	1,388,185	IBEF, March 2026
India hospital market	US\$135.3 bn in 2026, US\$202.5 bn by 2030	IBEF

Serviceable market, on the counted base. The 16,746 empanelled private hospitals are a register, not an estimate. One line each at the intended price of Rs 6,000 a month is **US\$13.7M a year**.

We do not publish a total addressable market. Standalone clinics are the larger part of Indian outpatient care and no official register we could find counts them. Deriving a number from the doctor count would need two ratios we cannot source, so the figure is left out rather than invented.

Converted at Rs 88 to the US dollar. The price is our intention and has never been charged to anyone; the base it multiplies is counted.

Revenue

Subscription per clinic line. Rs 6,000 a month including 500 answered minutes, then Rs 4 a minute. Intended pricing, not tested on a buyer.

Per verified chart. For clinics that would rather pay per outcome than per month, priced on charts where every critical field came out verified.

Where it grows. Reminders and recalls on the same line. Multi-clinic groups, where the desk view becomes the product. An audit export for insurers and accreditation, which is what the graded chart already is.

What it costs to serve. Platform minutes and one second transcription per call. The margin is set by call length, so a shorter, better intake is worth more than a longer one.

Competitors, and what is different here

Who	What their own site says they do
Assort Health	Voice agents across the care journey for US providers, at real scale: 250M+ patient interactions, 89% reduction in hold times. Confirms the action it is about to take, for example "can I confirm you for Thursday at 9:30".
Infinitus	Inbound and outbound healthcare calls for pharma, payors and providers. Enforces verbatim language where required, with a clinical team evaluating agents before, during and after calls.
Hyro	AI agents for healthcare call centres, scheduling and patient access.
Building it yourself	A voice API and a prompt. Fastest to start, and the artefact at the end is still the model's own account of the call.

The unique selling proposition. These are real companies with real deployments, ahead of this on scale, integrations and latency. Reading their public material in September 2026, we did not find one that hands the clinic a per field verdict drawn from a second decode of the same call, with the disagreements named. That artefact is what Twiceheard produces, and it is the part a clinic can be held to.

Descriptions above are quoted or paraphrased from each company's own public site, read in September 2026. The claim about what we did not find is a claim about our reading, not about what those products do internally.

Future prospects

Next. Hindi and Hinglish replies, not only Hindi understanding. Writing the graded chart into the clinic's own record system. A desk view across several clinics.

Scale. The verification is per call and holds no state between calls, so it scales with the voice platform underneath it. The cost per call is one extra transcription.

Beyond intake. Where we would take it next: pathology callbacks, discharge instructions, pharmacy orders. Anywhere a spoken value has to be right. None of that is built.

Why it holds. Models will keep getting better at hearing. None of that removes the need to prove, afterwards, that what was written down is what the person said.

Where it honestly stands

- The phone line, the browser call, the graded chart and the desk all work, in production, today.
- The demo deployment sends no text messages. The number a caller gives is never verified as theirs, so a public line that texts it is a way to text strangers at the operator's expense.
- The median caller gap is 3.4 s. The market sits between 1.7 s and 2.5 s. Almost all of ours is the platform's turn, and it is the main thing still wrong.
- One clinic, defined in code. There is no screen for adding another.
- No grievance contact is published, so the deployment is not lawful for live patient calls in India until a clinic publishes its own.

Every claim in this deck is checkable against the repository, and the ones that were not have been removed from it.